Band-1 Hadamard Order 8356 – Miyamoto (Paley-I Seed H_2088 q=2087 + C_2090) + Paley-II q=4177 IN SPK; Gap 2052 Still OPEN

H_2088 Paley-I PASS; C_2090 PASS; H_8356 Miyamoto PASS; H_8356 Paley-II q=4177 PASS -> IN SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8116_PaleyII`; `2026-09-16_Band1_Hadamard_7988_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8356_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2088 and 8356 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{2088} = Paley-I($q = 2087$) and symmetric Paley conference C_{2090} over GF(2089) ($q = 2089 \equiv 1 \pmod 4$), **and** classical Paley-II H_{8356} = Paley-II($q = 4177$) (prime $\equiv 1 \pmod 4$). Classical SPK for 8356: Paley-I **FAIL**s ($q = 8355 = 3 \cdot 5 \cdot 557$ composite); Paley-II **PASS**es ($q = 4177$ prime $\equiv 1 \pmod 4$); Kronecker **FAIL**s (4×2089). Therefore order **8356 is IN SPK via Paley-II** – Miyamoto alone never folds an order into SPK. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **8116 IN SPK via Paley-II** (seeds **2028+8116**). Prior SPK seeds **2028 / 8116 / 1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **2088** and **8356** folded into ONBOX/SPK when verified (8356 via Paley-II only).

Abstract

Finite constructions of orders **2088** and **8356**.

- H_{2088} = Paley-I ($q = 2087$) – preferred seed (Paley-II $q = 1043$ FAIL composite; Double FAIL; Kronecker 4×522 FAIL)
- C_{2090} = Paley conference over GF(2089), $q = 2089$
- H_{8356} = Miyamoto(H_{2088}, C_{2090})
- H_{8356} = Paley-II ($q = 4177$) – classical SPK path
- Paley-I for 8356: **SKIP** ($q = 8355$ composite)
- Kronecker for 8356: **SKIP** ($4 \cdot 2089$ not admissible)

SPK: 8356 IN SPK via Paley-II $q = 4177$. Seeds 2088 and 8356 enter SPK. Prior SPK **2028 / 8116 / 1996 / 1932 / 7604 / 7508 / 1876 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8356.py`

Results. overall_pass=true; units 137/137; wall 957.3944s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8356 = 4 \cdot 2089$; 2089 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{2088} and C_{2090} (Paley conference over prime field $q = 2089$).
- Seed: Paley-I $q = 2087$ (prime $\equiv 3 \pmod 4$) -> H_{2088} (preferred); Paley-II $q = 1043$ FAIL; Double FAIL; Kronecker 4×522 FAIL.
- SPK: Paley-I $q = 8355$ **FAIL** (composite $3 \cdot 5 \cdot 557$); Paley-II $q = 4177$ **PASS** (prime $\equiv 1 \pmod 4$); Kronecker **FAIL** -> **IN SPK via Paley-II**.
- Prior: 8116 IN SPK via Paley-II shipped; this ship $q = 2089 \rightarrow n = 8356$ (+ Paley-II SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_2088 Paley-I	Paley-I q=2087 DIRECT	0	6.0813
C_2090	Paley conf GF(2089) q=2089	0	5.6107
H_8356 Miyamoto	Miyamoto(H_2088 Paley-I, C_2090)	0	464.4725
H_8356 Paley-II	Paley-II q=4177 (classical SPK)	0	470.008
H_8356 Paley-I	SKIP q=8355=3*5*557	–	–
H_8356 Kronecker	SKIP 4*2089 not admissible	–	–

SPK closure note

Seeds include prior SPK ships through **8116** lineage (**2028** Paley-I; **8116** Paley-II; **1996** Paley-II; **1932** Paley-I; **7604** Paley-I; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/638 + verified **2088** (Paley-I) + verified **8356** (Paley-II q=4177); **exclude** Miyamoto-only bonuses. Known count 1668. Next constructive gap **2052** (OPEN).

Quantity	Value
8356 in SPK	**True** (Paley-II q=4177)
2088 in SPK	**True** (Paley-I q=2087)
2028 / 8116 / 1996 / 1932 / 7604 / 7508 in SPK	**True** (prior)
7988 / 7732 / 7492 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture * no Clay-closed * no finished-ToE * no RH / P!=NP / YM / Frankl / NS-solved / AC-solved * gap 2052 remains OPEN * finite constructions of 2088+8356 ONLY * Miyamoto alone never folds into SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_8356_miyamoto/`
Artifacts: `H2088_paleyI_pm1.npz`, `C2090_pm1.npz`, `H8356_miyamoto_pm1.npz`, `H8356_paleyII_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.